HAW HAMBURG

BACHELORTHESIS
Jon Doe

Life, the Universe and Everything

FACULTY OF COMPUTER SCIENCE AND ENGINEERING
Department of Information and Electrical Engineering

Fakultät Technik und Informatik
Department Informations- und Elektrotechnik

HAMBURG UNIVERSITY
OF APPLIED SCIENCES

Hochschule für Angewandte
Wissenschaften Hamburg

Jon Doe

Life, the Universe and Everything

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Abstract

Arthur Dents travel to a new future ...

Jon Doe

Thema der Arbeit

Das Leben, das Universum und der ganze Rest

Stichworte

Leben, Universum, Alles

Kurzzusammenfassung

Arthur Dents Reise in eine neue Zukunft ...

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1 Introduction

This is a placeholder citation so that BibTeX has at least one reference to process [1].

2 Theory

3 Requirements Analysis

This chapter establishes the objectives and requirements of the present work. To provide a clear specification for the developed solution, the requirements are derived systematically through a step-by-step analysis based on the following.

3.1 Research Aim and Questions

The overall aim of this thesis is two-fold. It is to first understand the influence of power amplifier nonlinearity on the SSR-X [JV1.1]power amplifier chain, and secondly, to counteract these effects using digital predistortion. These unwanted nonlinear effects appear beginning at a certain input power threshold. Digital predistortion can counteract these effects, enabling the power amplifier to run closer to saturation without violating linearity limits. This keeps the radar spectrally compliant and enables the radar to accommodate higher input power and consequently higher peak output power. This increased headroom can be used either to increase radar detection range or detect targets with a smaller cross section at the same range – which is a direct performance gain. Mitigating nonlinearity effects may also aid in shrinking the masking region around strong targets, recovering the ability to detect weaker targets close in range to them. Moreover, if the masking region level reaches a certain threshold, false targets may emerge – misleading navigators. Mitigating such effects is paramount to improving radar performance and reliability. Specific research questions pertinent to the overall objective are outlined below:

- **RQ1:** What are the effects of power amplifier non-linearity on the power amplifier chain and performance?
- **RQ2:** What existing models best model the power amplifier non-linearity effects on the power amplifier chain?

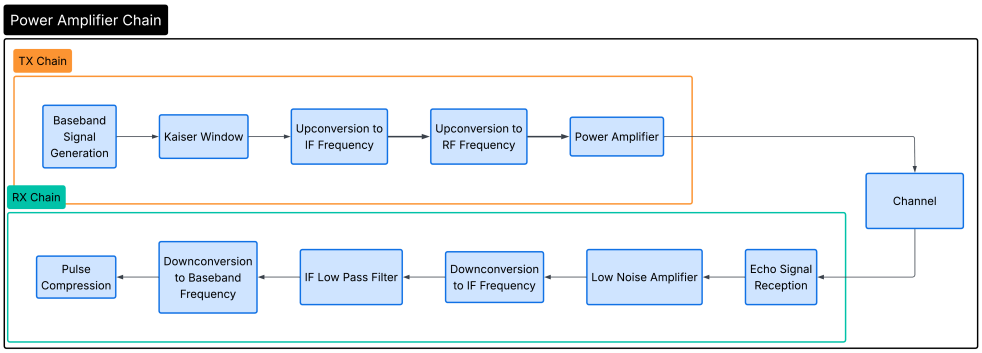


Figure 3.1: SSR-X Power Amplifier Chain

- **RQ3:** What digital predistortion techniques exist to counteract the non-linear effects on the power amplifier chain?
- **RQ4:** To what extent can digital predistortion counteract the non-linear effects on the power amplifier chain and improve performance?

3.2 System Environment

This section describes the environment of our system. Our system is confined to the power amplifier chain of the SSR-X radar transceiver[JV2.1], shown in Figure 3.1. A radar pulse signal is generated in the transmit chain and then input into the power amplifier for signal amplification before transmission. The echo of this pulse is then captured by the receive chain after some delay and attenuation. Pulse compression is performed at the end of the signal chain to increase SNR, increase range resolution, and determine range of a detected target. The thesis work focuses on investigating the nonlinear effects the power amplifier has on the chain and finding a suitable digital predistortion solution to correct for these effects.

3.3 Stakeholders

The stakeholders of this thesis are those who influence the direction of work and benefit from its results. They are outlined below along with their interests:

- **NACOS Marine Radar Developers:** The engineers at NACOS Marine, who are responsible for the maintenance and development of the radar system, benefit from the results of this work, as it will provide insight on what the effects of power amplifier nonlinearity have on the radar system and its performance. It will help them understand the extent of these effects and how they may be detrimental to the overall radar system performance. The work may also provide potential solutions to these issues – guiding engineers in improving the quality and performance of the product they are developing.
- **NACOS Marine Product Management:** The product management department at NACOS Marine may benefit from this work, as it may improve the performance of the product they are selling. Higher performance – increased target range or detection of smaller targets – may open new market potential, increasing sales and revenue of the product.
- **International Electrotechnical Commission (IEC) Technical Committee:** According to standard MSC. 192/5.1.1 and MSC. 192/5.1.2 within the International Standard IEC 62388, X-band maritime radar shall transmit within the confines of the allocated band 9.2-9.5 GHz. [JV3.1]IEC are interested in the compliance of this requirement. Power amplifier nonlinear effects may result in unwanted energy spreading outside of the confined frequency band – violating this requirement. This thesis work may present solutions to countermand these effects, ensuring spectral compliance with the standard.
- **Radar System Users:** Users of the radar system, namely crew members responsible for navigating their vessel at sea, are interested in the results of this thesis, because it will potentially improve performance and reliability of the radar image used for navigating safely. The solutions presented in this thesis may aid in reducing masking around strong targets – enabling weaker targets within their vicinity to be distinctly detected. There is also the risk of the emergence of false targets – which this work may potentially mitigate. This helps crews to be sure that they can trust their navigation instruments. Performance gains, such as increased range and detection of smaller targets, may also help crews navigate more safely to avoid collision.
- **HAW Hamburg Lecturers and Students:** Should the results of this thesis be made public and accessible to the university, lecturers and students may benefit from it. Firstly, lecturers may be interested in its content for instructive reasons to

teach students concepts of radar, signal processing, or general structure of scientific thesis work. Students may benefit from the work as a template for their own theses or for the aforementioned learning content.

- **The Student:** The student, the author of this thesis, is interested in learning about the technicalities of radar development and deepening their understanding of concepts in signal processing. This thesis work provides the student the opportunity to gain practical experience in these aspects.

3.4 Objectives Analysis

The overarching objective of this work is to determine what real effects power amplifier nonlinearity has on the power amplifier chain and appropriate predistortion techniques to amend such effects for performance gain and spectral compliance. To better solidify how this can be achieved, it is divided into several detailed objectives below.

3.4.1 Generic Modelling of the Power Amplifier Chain

The first objective is to develop a power amplifier chain simulation that can be used as a controlled platform for nonlinearity and digital predistortion experiments. This simulation serves as a generic model to understand the general behaviour of the power amplifier chain and apply different parameter values and view how this affects the outputs. Within such a simulation, various power amplifier nonlinearity models from literature can be dynamically applied and their effects investigated. On that note, a simulation will also demonstrate what the outputs of an ideal system would be, void of nonlinear effects. Furthermore, the simulation forms the basis for experimenting with different digital predistortion algorithms.

To best understand the power amplifier chain behaviour, it stands to reason for the simulation to simulate the main stages of the chain and pulses generated by the real system in its standard operating mode. The pulses comprise a short, medium, and long pulse – where short and medium are used for ranges 0.75 to 1.5NM[JV4.1], medium and long are used for 24 to 48NM, and the combination of all three is used for ranges 3 to 24NM. These pulses are generated in the real system's FPGA as baseband signals. They are then windowed, converted to analogue signals, and upconverted to IF and RF

frequencies preceding amplification for transmission. This forms the transmit chain. Each pulse would return an attenuated and time delayed echo then passed through the receive chain of the system. In the receive chain, the echoes are amplified, down converted back to IF and baseband frequencies, and low pass filtered at the IF frequency. The final stage of the chain entails pulse compression to maximise SNR and detect after which delay time the echo returns. Each of the above stages act as primary components forming the entire power amplifier chain.

Furthermore, the simulation should produce plots that can be used to quantitively analyse power amplifier nonlinearity effects. Plots relevant to this topic are namely spectra of the various signals across the chain, AM-AM and AM-PM curves of the power amplifier, and the pulse compression output. Therefore, it stands to reason for the model to generate said plots. To evaluate overall performance, relevant plots should include quantitative values for KPIs pertinent to known nonlinearity effects – these primarily encompass signal magnitude, side lobe level, AM-AM linearisation, and AM-PM phase deviation. To elaborate, spectrum plots should measure main lobe peak amplitude and adjacent channel power ratio (ACPR) as KPIs to determine effects on signal magnitude and spectral compliance. The pulse compression output should detail KPIs of peak side lobe level (PSL), integrated side lobe level (ISL), and main lobe peak amplitude – as these are metrics for signal magnitude and the masking region. AM-AM curves should measurably highlight the linear range extension as a KPI for linearisation effect, whereas total phase deviation should constitute the KPI for AM-PM curves.

This objective is achieved if the simulation is developed and supports all three pulses and the experimentation of power amplifier nonlinearity and digital predistortion models – involving the generation of relevant measurement plots with quantitative values for KPIs.

3.4.2 Precise Modelling of the Power Amplifier Chain

Following completion of the generic power amplifier chain model, the second objective is to adjust the parameters of the simulation to best model the actual power amplifier chain hardware. This will allow experiments on the nonlinearities and digital predistortion to be relevant to the real-world system. This will result in understanding the actual effects nonlinearity has on the power amplifier chain hardware and how it reduces performance.

This objective involves performing lab measurements on the power amplifier chain hardware to obtain the same plots generated by the simulation. This means the measurement step should endeavour to produce the spectrum plots of the chain's signals at each stage, AM-AM and AM-PM power amplifier curves, and the pulse compression result. Analogous to the simulation, KPIs – signal magnitude, side lobe level, AM-AM nonlinear deviation, and AM-PM phase deviation – should be obtained as quantitative values from these plots.

When all measurement plots and values are obtained, simulation parameters can be finetuned to emulate the hardware as closely as possible. This includes fitting the configuration of the short, medium, and long pulses and power amplifier nonlinear behaviour amongst other potential discrepancies across each stage in the chain. The simulation should thus ideally include a variety of power amplifier nonlinearity models, so a comparison can be made between models to determine which best matches the real system. This is conducive to understanding the actual nonlinearity effects on the hardware and deducing its effects on performance.

This objective is achieved when all appropriate measurements are obtained and the simulation finetuned with the best matching power amplifier model chosen to precisely emulate the power amplifier chain hardware.

3.4.3 Implementation of Digital Predistortion

Once concrete nonlinear effects are determined from the precise model, the final objective involves implementing digital predistortion to counteract these effects and improve performance. This would involve the application of digital predistortion algorithms in the simulation, where the counter-effects on nonlinearity can be observed in the generated simulation results. To compare and evaluate results, it stands to reason to generate three separate sets of results – one ideal, one with nonlinear effects modelled, and one with digital predistortion applied. Values of the curves' KPIs – signal magnitude, side lobe level, AM-AM nonlinear deviation, and AM-PM phase deviation – can then be directly compared. This aids in providing a metric to quantitively evaluate the extent of performance gain digital predistortion introduces. This same comparison and evaluation could potentially also be performed on the real power amplifier chain hardware by implementing the simulated digital predistortion algorithm in its firmware.

This objective is achieved when a digital predistortion algorithm is implemented in the simulation (and possibly in the firmware) and ideal, nonlinear, and predistorted results are compared with subsequent performance gain evaluated.

3.5 Requirements Overview

Based on the system environment and analysis of stakeholders and objectives, the following functional and non-functional requirements are derived. Functional requirements describe behaviours that the system must provide, whereas non-functional requirements describe that quality constraints the system must adhere to e. g. performance, reliability and usability. These requirements must be concrete and testable to determine whether the developed system has truly fulfilled them. The requirements directly impact decisions on system design and final evaluation.

3.5.1 Functional Requirements

FR1: The power amplifier chain simulation shall include the following modular stages for complete modelling of the real system:

- Baseband signal generation
- Baseband signal windowing
- Up conversion to IF frequency
- Up conversion to RF frequency
- Transmit chain amplification (power amplifier)
- Channel effects (delay and attenuation)
- Receive chain amplification
- Down conversion to IF frequency
- IF frequency low pass filter
- Down conversion to Baseband frequency
- Pulse compression

FR2: The power amplifier chain simulation shall include the short, medium, and long pulses generated by the real system respectively.

FR3: The power amplifier chain simulation shall apply at least two known power amplifier nonlinearity models for comparison.

FR4: The power amplifier chain simulation shall apply at least one known digital predistortion algorithm for evaluation.

FR5: For each transmit pulse type (short, medium, and long), the power amplifier chain simulation shall generate analysis plots that overlay the ideal, nonlinear power amplifier, and predistorted power amplifier cases for direct comparison. The generated plots shall include:

- Spectrum of respective transmit pulse at Baseband frequency pre windowing
- Spectrum of respective transmit pulse at Baseband frequency post windowing
- Spectrum of respective transmit pulse at IF frequency
- Spectrum of respective transmit pulse at RF frequency pre amplification
- Spectrum of respective transmit pulse at RF frequency post amplification
- Spectrum of the respective receive echo upon reception
- Spectrum of the respective receive echo post amplification
- Spectrum of the respective receive echo at IF frequency pre filtering
- Spectrum of the respective receive echo at IF frequency post filtering
- Spectrum of the respective receive echo at baseband frequency
- Pulse compression output
- Power amplifier AM-AM curve
- Power amplifier AM-PM curve

FR6: Generated power amplifier chain simulation spectrum plots shall include quantitative values for the following KPIs for evaluation:

- Main lobe peak amplitude

- Adjacent channel power ratio (ACPR)

FR7: Generated power amplifier chain simulation pulse compression plots shall include quantitative values for the following KPIs for evaluation:

- Main lobe peak amplitude
- Peak side lobe level (PSL)
- Integrated side lobe level (ISL)

FR8: Generated power amplifier chain simulation AM-AM plots shall include quantitative values for the KPI of linear range extension for evaluation.

FR9: Generated power amplifier chain simulation AM-PM plots shall include quantitative values for the KPI of total phase deviation for evaluation.

FR10: Measurements on the power amplifier chain hardware shall produce the following plots respectively for the short, medium, and long pulse for simulation finetuning:

- Spectrum of respective transmit pulse at Baseband frequency pre windowing
- Spectrum of respective transmit pulse at Baseband frequency post windowing
- Spectrum of respective transmit pulse at IF frequency
- Spectrum of respective transmit pulse at RF frequency pre amplification
- Spectrum of respective transmit pulse at RF frequency post amplification
- Spectrum of the respective receive echo upon reception
- Spectrum of the respective receive echo post amplification
- Spectrum of the respective receive echo at IF frequency pre filtering
- Spectrum of the respective receive echo at IF frequency post filtering
- Spectrum of the respective receive echo at baseband frequency
- Pulse compression output
- Power amplifier AM-AM curve
- Power amplifier AM-PM curve

FR11: Measured power amplifier chain hardware spectrum plots shall include quantitative values for the following KPIs for simulation finetuning:

- Main lobe peak amplitude
- Adjacent channel power ratio (ACPR)

FR12: Measured power amplifier chain hardware pulse compression plots shall include quantitative values for the following KPIs for simulation finetuning:

- Main lobe peak amplitude
- Peak side lobe level (PSL)
- Integrated side lobe level (ISL)

FR13: Measured power amplifier chain hardware AM-AM plots shall include quantitative values for the KPI of linear range extension for simulation finetuning.

FR14: Measured power amplifier chain hardware AM-PM plots shall include quantitative values for the KPI of total phase deviation for simulation finetuning.

3.5.2 Non-Functional Requirements

NFR1: The power amplifier chain simulation shall be developed in Python to ease the learning curve and development speed for the student.

NFR2: The power amplifier chain simulation shall be developed modularly, so that it can be easily maintained and extended.

NFR3: The documentation and implementation of the work shall be well structured and comprehensible to ensure future developers or researchers can inspect, reproduce, or extend the work. This entails clear project structure, fully documented project components, and repletely annotated results.

4 Concept and Design

5 Implementation

6 Evaluation

7 Conclusion and Outlook

Bibliography

[1] Adams, D.: *Douglas Adams Trilogy/Hitchhiker's Guide to Galaxy/Restaurant at the End of the Universe/Life, the Universe and Everything*. Random House Incorporated, 1995. – URL https://books.google.de/books?id=bMWkPQAACAAJ. – ISBN 9780345397041

A Appendix

Declaration

I declare that this Bachelor Thesis has been completed by myself independently without outside help and only the defined sources and study aids were used.

City	Date	Signature